Customer Requirements Analysis Report

Autumn 2025

Topic: Simulating operations of Wholesale Club (HyperMarket)

Group: 16

List of users

The following list of users should be updated automatically. Do not edit it manually.
You can press the “reload” button on the left to update it

2221338 – Nafiu Islam
- User 1: Cashier
- User 2: Inventory Manager

2312150 – Mubassir Mohi Mojumder
- User 3:Store Manager
- User 4: Membership Manager

2310106 – Rubayet Rahman Rahad
- User 5: Delivery Coordinator
- User 6: Procurement Officer

2312026 – Tanjil Sarkar
- User 7: HR Manager
- User 8: Customer Support Staff

Instructions

- The group leader will Download/Create a copy of the document, then share with other group members.
    - Also give viewer access to your course instructor (asif.mahmood@iub.edu.bd)
- Each member will be responsible for completing his/her sections of the document
- Make a PDF of the document and submit using the given google form.
    - Do not forget to change the file name according to the instructions given in the form.

How to complete this report

The parts you have to complete/update have been marked in RED.

- On page 1, add your project topic (e.g. Simulating operations of …) and your group number.
    - Do NOT modify the list of users. It will be updated automatically
- Add your Name and ID in place of "1234567 – Name of group member X" on page 3 onwards
    - Update the footer accordingly
- Add names of users 1-8
- Check the first page and verify that group member names and users have been updated.
- Add description of goals 1-8 of each user
- Add workflow tables of goals 1-8 of each user
    - Add table rows if necessary

After you have finished updating the report, change font colors from RED to BLACK.

Possible workflow event type:

- UIE - - user input to trigger event
- UID - - user input to be considered as data
- OP – display content (output)
- VL – validation check
- VR – verification check
- DP – fetching data from file system and process it to get some calculated outcome (data processing)

2221338 – Nafiu Islam

User 1: Cashier

User 1 Goal 1

Process Customer Checkout: The cashier starts checkout, enters the product ID, the system calculates the total price, and then checks if membership discounts apply.

Workflow:

Event #	Description	Type
Event 1	The cashier clicks the **“Start Checkout”** button to begin the billing process.	UIE
Event 2	The cashier enters the product ID **“fish1”** into the system.	UID
Event 3	The system processes the entered ID and calculates the total price after clicks **“Add Product”**	DP
Event 4	After clicking **“Check Membership”** the system checks if the customer has a membership discount that can be applied.	VL

User 1 Goal 2

Apply Membership Discount: This goal allows the cashier to verify a customer’s membership and apply a discount to their total bill.

Workflow:

Event #	Description	Type
Event 1	The cashier clicks the **“Apply Membership”** button to begin the discount process.	UIE
Event 2	The cashier enters the customer’s membership ID **“111”** into the system.	UID
Event 3	The system checks if the entered membership ID is valid after clicking the **“Verify”** button and it shows whether it is **“positive”** or **“Invalid ID”.**	VR
Event 4	When click on the **“Recalculate Total With Discount”** button the system recalculates the total price by applying the membership discount	DP
Event 5	The system shows the updated bill with the applied discount.	OP

User 1 Goal 3:

Process Payment: This goal allows the cashier to take the customer’s payment, validate it, and confirm successful completion of the transaction.

Workflow:

Event #	Description	Type
Event 1	The cashier selects the preferred payment method (Cash or Card) from the system.	UIE
Event 2	The cashier enters the payment amount **“50”** into the payment field.	UID

Event 3	The system validates the payment information to ensure it is correct and acceptable.	VL
Event 4	The system displays a message confirming that the **"Payment Complete"**	OP

User 1 Goal 4

Handle Returns/Refunds: This goal allows the cashier to verify a customer's return request, check eligibility, calculate refund amount, and confirm the refund process.

Workflow:

Event #	Description	Type
Event 1	The cashier selects the **"Return Item"** option to start the return/refund process.	UIE
Event 2	The cashier enters the product ID **"fish1"** and bill number **"f111"** provided by the customer.	UID
Event 3	The system verifies whether the product is eligible for a return or refund and show message **"Positive"** or **"Not Eligible"**	VR
Event 4	If eligible, the system processes and calculates the correct refund amount **"100 taka" and** if not eligible **"0"**	DP
Event 5	The system displays the refund confirmation **"Confirm Refund"** or denial message **"Refund Denied"** to the cashier.	OP

User 1 Goal 5

Generate Daily Sales Report: This goal enables the cashier to generate the daily sales report by retrieving the day's transaction data, calculating total sales, and displaying the final summarized report.

Workflow:

Event #	Description	Type
Event 1	The cashier selects the **"Generate Report"** option to begin generating the daily sales report.	UIE
Event 2	The system retrieves and displays **"Total Transaction: 5"** data recorded for the current day.	DP
Event 3	The system summarizes the total transactions and calculates the net sales amount **"500 taka"** and saves this into a `"totalcash.txt"` file.	DP
Event 4	The system displays the final sales report and shows **"Report Generate Successfully"** to the cashier.	OP

User 1 Goal 6

Balance Cash Drawer: This goal allows the cashier to verify the end-of-shift cash balance by entering the actual counted cash, comparing it with the system's expected amount, identifying any differences, and displaying the final balance status.

Workflow:

Event #	Description	Type

Event 1	The cashier selects the **“End Shift”** option to start the cash balancing process.	UIE
Event 2	The cashier enters the actual amount **“100”** of cash counted in the drawer and click the **“Submitted Cash Count”** button.	UID
Event 3	The system compares the entered cash amount **“100”** with the expected amount **“100”** from sales records.	DP
Event 4	The system checks whether there is any overage or shortage in the cash drawer.	VL
Event 5	The system displays the final cash balance status **“Cash Drawer Balanced”** or **“Mismatch Found”** .	OP

User 1 Goal 7

Queue Management: This goal allows the cashier to manage the customer queue by recording each served customer and showing when the system is ready for the next person.

Workflow:

Event #	Description	Type
Event 1	The cashier clicks the **“Next Customer”** button to move to the next person in the queue.	UIE
Event 2	The system logs the transaction by increasing the customer count and noting that a customer was served.	DP
Event 3	The system displays a **“Ready for Next Customer”** status for the cashier.	OP

User 1 Goal 8

Flag Suspicious Activity: This goal allows the cashier to report any suspicious transaction by writing a reason, saving it into a flagged list, and notifying the manager.

Workflow:

Event #	Description	Type
Event 1	The cashier selects a **“Suspicious Report”**	UIE
Event 2	The cashier enters the reason or details of the suspicious activity in the text field and clicks the **“Submit Report”** button .	UID
Event 3	The system saves the suspicious activity record into a `suspiciousactivity.txt` file .	DP
Event 4	The system displays **“Record Saved”** and **“Notified”** .	OP

User 2: Inventory Manager

User 2 Goal 1

Monitor Stock Levels: This goal allows the inventory manager to check the current stock of any product by entering a product name and viewing its available quantity.

Workflow:

Event #	Description	Type
Event 1	Inventory Manager clicks **“Check Inventory”** to start the stock lookup.	UIE
Event 2	The system fetches stock data from the database based on the entered product name **“fish”** .	DP

Event 3	The system displays the available quantity **“5”** of the product in the quantity field.	OP

User 2 Goal 2

Raise Purchase Request: This goal allows the inventory manager to request new stock by entering a product ID and the required quantity. The system validates the inputs and then submits the purchase request.

Workflow:

Event #	Description	Type
Event 1	Inventory Manager clicks **“New Purchase Request”** to start the request process.	UIE
Event 2	The user enters Product ID and Required Quantity into the input fields.	UID
Event 3	The system processes and checks if the request matches the allowed condition (`productId = fish1` and `quantity = 10`).	DP
Event 4	System displays confirmation message: • **“Request Successfully Submitted”** • Or **“Request not Submitted”**	OP

User 2 Goal 3

Receive Supplier Delivery: This goal allows the inventory manager to verify an incoming supplier shipment using the invoice number and then update stock quantities based on the delivered product list.

Workflow:

Event #	Description	Type
Event 1	Inventory Manager selects **“Receive Delivery”** to open the delivery page.	UIE
Event 2	User enter Invoice Number **“111”** and Product List .	UID
Event 3	After clicking **“Verify With Purchase Order”** System validates invoice number.	VL
Event 4	The system displays **“Product Verified for Invoice:111” .**	OP
Event 5	The system updates stock quantities based on product list.	DP
Event 6	The system displays updated inventory when clicking on the **“Update Stock Quantities”** and saves this data to the `"Receive_Supplier_Delivery.txt"` file.	OP

User 2 Goal 4

Update Damaged/Expired Goods: This goal allows the inventory manager to verify stock, record damaged or expired items, deduct the quantity from available stock, and save the record for tracking.

Workflow:

Event #	Description	Type
Event 1	Select **“Update Damaged/Expired Goods”**	UIE
Event 2	Enter **Product ID**, **Product Name**, **Damaged Quantity**, and **Reason.**	UID
Event 3	System verifies current stock **(displays “Stock Verified”)**	VR
Event 4	System validates entered quantity (checks empty, non-number, or too large)	VL
Event 5	The system deducts stock and saves all the information into the `"damaged goods.txt"` file.	DP
Event 6	The system displays **“Stock Deducted: Updated Stock: ”** .	OP

User 2 Goal 5

Forecast Demand: This goal allows the inventory manager to input past sales data and generate a simple demand forecast based on a fixed prediction logic.

Workflow:

Event #	Description	Type
Event 1	Select **“Forecast Demand”**	UIE
Event 2	Enter past sales data **“10,20,30,40”**	DP
Event 3	The system processes each number and applies prediction logic **(adds +40 to each value).**	DP
Event 4	The system displays the generated demand forecast **“50,60,70,80”** in the forecast report area.	OP

User 2 Goal 6

Generate Stock Report: This goal allows the Inventory Manager to input current inventory data manually and generate a stock summary indicating low-stock and overstock items.

Workflow:

Event #	Description	Type
Event 1	Click **“Generate Stock Report” .**	UIE
Event 2	Enter current inventory data **“Fish:5, Rice:120, Meat:50”**	DP
Event 3	The system processes each record and identifies low-stock or overstock items. “**Stock Report:** **Fish: 5 (Low Stock)** **Rice: 120 (Overstock)** **Meat: 50”**	DP
Event 4	System displays the generated stock report in the summary area	OP

User 2 Goal 7

Organize Warehouse: This goal allows the Inventory Manager to assign a storage shelf/location to a product and verify if the location is available.

Workflow:

Event #	Description	Type
Event 1	Select **“Organize Warehouse”**	UIE
Event 2	Enter product ID **“fish1”** & shelf location **“A1”**	UID
Event 3	System checks and confirms location availability by clicking “**Validate Location” .**	VL
Event 4	When clicking on the **“Update Placement”** button, the system updates the product’s storage placement.	DP
Event 5	Display confirmation of updated storage location **“Product fish1 placed in A1” .**	OP

User 2 Goal 8

Handle Supplier Returns: This goal allows the Inventory Manager to verify if a product can be returned to the supplier and then process the return.

Workflow:

Event #	Description	Type
Event 1	Select **“Handle Supplier Returns”**	UIE
Event 2	Enter product ID & quantity	UID
Event 3	By clicking on the **“Verify Return”** button system verifies if the product is eligible for return.	VR
Event 4	System processes the return and marks it as returned after clicking **“Process Return Button”**	DP
Event 5	Display return confirmation message.	OP

2312150 – Mubassir Mohi Mojumder

User 3:Store Manager

User 3 Goal 1

Daily Operations

Workflow:

Event #	Description	Type
Event 1	Click “Daily Operation”	UIE
Event 2	Enter daily sales from cashier(Physical report) , low stock info from inventory manager(Physical report) , and active staff count	UID
Event 3	Save daily operation in a file	DP
Event 4	Display confirmation	OP

User 3 Goal 2

Approve Purchase Order

Workflow:

Event #	Description	Type
Event 1	Click “Approve request”	UIE
Event 2	Enter product ID and product details from Inventory Manager (Physical report)	UID
Event 3	Update status to Approve/Reject	UIE
Event 4	Save decision for Procurement Officer	DP
Event 5	Show confirmation message	OP

User 3 Goal 3

Assign Staff Shifts

Workflow:

Event #	Description	Type
Event 1	Select “Assign Shift”	UIE
Event 2	Enter staff ID (Physical report), shift date, and shift time	UID
Event 3	Save to schedule database	DP
Event 4	Save shift record	OP

User 3 Goal 4

Sales Performance

Workflow:

Event #	Description	Type
Event 1	Select “Sales Report”	UIE
Event 2	Load sales data from cashier files	DP
Event 3	Show total sales	DP
Event 4	Display sales report	OP

User 3 Goal 5

Staff Complaints

Workflow:

Event #	Description	Type
Event 1	Click “Staff Complaints”	UIE
Event 2	Enter staff ID (Physical report) and complaint details	UID
Event 4	Save complaint file	DP
Event 5	Display confirmation message	OP

User 3 Goal 6

Approve Discounts & Promotions

Workflow:

Event #	Description	Type
Event 1	Click “Promotions & Discount”	UIE
Event 2	Enter product title/name (Physical report from Inventory Manager)	UID
Event 3	Enter previous price and new discount/promo price	UID
Event 4	Save promotion to file	DP
Event 5	Show confirmation	OP

User 3 Goal 7

Top Management Reports

Workflow:

Event #	Description	Type
Event 1	Click “Top Management Report”	UIE
Event 2	Load daily operation records	DP
Event 3	Summarize into report	DP
Event 4	Display final management report	OP

User 3 Goal 8

Assign Staff Task

Workflow:

Event #	Description	Type
Event 1	Click “Assign Task”	UIE
Event 2	Enter staff ID and select date	UID
Event 3	Select task area	UID
Event 4	Save task assignment	DP
Event 5	Show confirmation	OP

User 4: Membership Manager

User 4 Goal 1

Register New Membership

Workflow:

Event #	Description	Type
Event 1	Click “New Membership”	UIE
Event 2	Enter member name and membership type	UID
Event 3	Set valid-till date	UID
Event 4	Click “Generate ID” to create ID	UIE
Event 5	Save new membership record	DP
Event 6	Show confirmation	OP

User 4 Goal 2

Renew Memberships

Workflow:

Event #	Description	Type
Event 1	Click “Renew Membership”	UIE
Event 2	Enter member ID	UID
Event 3	Verify existing ID	VR
Event 4	Extend Validity & update records	UID
Event 5	Confirmation message	OP

User 4 Goal 3

Cancel Memberships

Workflow:

Event #	Description	Type
Event 1	Click “Cancel Membership”	UIE
Event 2	Enter member ID	UID
Event 3	Verify membership ID	VR
Event 4	Display member info	OP
Event 5	Click “Cancel Membership”	UIE
Event 6	Remove member record	DP
Event 7	Show confirmation	OP

User 4 Goal 4

Handle Lost Card Requests

Workflow:

Event #	Description	Type
Event 1	Click “Lost Card”	UIE
Event 2	Enter member ID	UID
Event 3	Verify identity	VR
Event 4	Generate card ID	DP
Event 5	Save updated ID	DP
Event 6	Show confirmation	OP

User 4 Goal 5

Update Membership Benefits

Workflow:

Event #	Description	Type
Event 1	Click “Membership Benefits”	UIE
Event 2	Enter member ID	UID
Event 3	Validate ID	VR
Event 4	Show current membership status	OP
Event 5	Update status	UID
Event 6	Save updated information	DP
Event 7	Display confirmation	OP

User 4 Goal 6

Generate Membership Reports

Workflow:

Event #	Description	Type
Event 1	Click “Membership Reports”	UIE
Event 2	Load membership records	DP
Event 3	Display records in table	OP
Event 4	Click “Generate PDF”	UIE
Event 5	Create and save PDF	DP
Event 6	Display confirmation	OP

User 4 Goal 7

Membership Info

Workflow:

Event #	Description	Type
Event 1	Click “Membership Info”	UIE
Event 2	Enter member ID	UID
Event 3	Click “Show Info”	UIE
Event 4	Load member record from file	DP
Event 5	Display data in table view	OP

User 4 Goal 8

Membership Issues

Workflow:

Event #	Description	Type
Event 1	Click “Membership Issues”	UIE
Event 2	Click “Load Issue”	UIE
Event 3	Read issue from customer support data	DP
Event 4	Display issue in text area	OP

2310106 – Rubayet Rahman Rahad

User 5: Delivery Coordinator

User 5 Goal 1

Assign Delivery Tasks

Workflow:

Event #	Description	Type
Event 1	User Clicks “Assign Delivery Tasks” from Delivery Dashboard	UIE
Event 2	User enters Staff ID, Order ID, and Delivery Date (from a manually given Sheet/paper of Confirmed Orders)	UID
Event 3	System validates inputs ,checks fields are not empty , checks if Order ID was already assigned	VL
Event 4	Saves the record in system	DP
Event 5	System shows “Delivery Assigned Successfully!” and clears input fields for next entry	OP

User 5 Goal 2

Track Delivery

Workflow:

Event #	Description	Type
Event 1	User clicks “Track Delivery” from the Delivery Dashboard	UIE
Event 2	User enters Order ID into the input field	UID
Event 3	System validates input , checks if Order ID field is empty , if empty, shows “Please enter an Order ID.”	VL
Event 4	System checks delivery status whether it’s delivered or In process	DP
Event 5	System displays appropriate status message to the user	OP

User 5 Goal 3

Confirm Completed Deliveries

Workflow:

Event #	Description	Type
Event 1	User Click “Confirm Delivery” from Delivery Dashboard.	UIE
Event 2	The user enters an Order ID into the input field and clicks Confirm Delivery.	UID
Event 3	System validates input	VL
Event 4	System checks if the order ID is assigned to any staff	DP
Event 5	Show confirmation message	OP

User 5 Goal 4

Report Returned Order Problem

Workflow:

Event #	Description	Type
Event 1	User clicks “Report Returned Order Problem” from the Delivery Dashboard	UIE
Event 2	Type order ID and problem details	UID
Event 3	Check if info is complete	VL
Event 4	System checks to verify that the order was already delivered:	DP
Event 5	System successfully reports the problem	OP

User 5 Goal 5

Manage Delivery Schedule

Workflow:

Event #	Description	Type
Event 1	User clicks “Manage Delivery Schedule” from the dashboard	UIE
Event 2	User inputs an Order ID and clicks Search. System validates input and checks if the order exists	UID/VL
Event 3	If the order is found, system displays previous delivery date	DP / OP
Event 4	User selects a new date and clicks “Update Date”	UIE
Event 5	System updates the delivery date in the file and confirms success	DP/OP

User 5 Goal 6

Generate Delivery Report

Workflow:

Event #	Description	Type
Event 1	User clicks “Generate Report” front the Dashboard	UIE
Event 2	System collects delivery data	DP
Event 3	System determines delivery status for each order and prepares report data	DP
Event 4	System displays the report in the TableView and shows success message	OP
Event 5	User clicks “Download PDF”	UIE

User 5 Goal 7

Verify Returned Orders

Workflow:

Event #	Description	Type
Event 1	User clicks “Verify Returned Orders” from Delivery Dashboard	UIE
Event 2	User enters an Order ID and clicks Verify	UIE/UID
Event 3	System collects all return problems from Return Problem Reports goal	DP
Event 4	System filters return problems based on entered Order ID	DP

Event 5	System displays the filtered return problems in the TextArea and updates the status label	OP

User 5 Goal 8

Re-Delivery Request

Workflow:

Event #	Description	Type
Event 1	User clicks “Re-Delivery Request” from Delivery Dashboard	UIE
Event 2	Enter Order ID and reason for re-delivery, then click Save Request	UID/UIE
Event 3	System verifies order status from Assigned Deliveries and Confirmed Orders goals	DP
Event 4	System saves re-delivery request under Re-Delivery Requests goal and removes order from Confirmed Orders	DP
Event 5	Display confirmation or error message	OP

User 6: Procurement Officer

User 6 Goal 1

Review Purchase Request

Workflow:

Event #	**Description**	**Type**
Event 1	Click REview Purchase Request from Procurement Dashboard	UIE
Event 2	User clicks Show Requests button	UIE
Event 3	Fetch approved purchase request list from Store manager	DP
Event 4	System displays all purchase requests in the TextArea	OP

User 6 Goal 2

Create Supplier Order

Workflow:

Event #	**Description**	**Type**
Event 1	Click Create Supplier Order from Procurement Dashboard	UIE
Event 2	Enter supplier ID, product, quantity, and due date; optionally generate Order ID(gets supplier id from manually given sheet/paper)	UID
Event 3	Click Save Order button	UIE
Event 4	System validates input and saves order under Supplier Orders goal	DP
Event 5	Display confirmation message	OP

User 6 Goal 3

Approve Supplier Quotation

Workflow:

Event #	Description	Type
Event 1	Click Approve Supplier Quotation from Procurement Dashboard	UIE
Event 2	Enter quotation details manually from supplier document	UID
Event 3	Click Approve or Reject button	UIE
Event 4	System validates input and saves the decision	DP
Event 5	Display success or error message	OP

User 6 Goal 4

Resolve Issues with Supplier

Workflow:

Event #	Description	Type
Event 1	Click Resolve Supplier Issues from Procurement Dashboard	UIE
Event 2	Load damaged product list from Inventory Manager	UID
Event 3	Display damaged products in table	OP
Event 4	Enter remarks and click Send Replacement Request	UID/UIE
Event 5	System processes replacement request and shows confirmation	OP

User 6 Goal 5

Manage Supplier Records

Workflow:

Event #	Description	Type
Event 1	Click Manage Supplier Records from Procurement Dashboard	UIE
Event 2	Enter supplier ID, name, contact, and address (from manual supplier sheet)	UID
Event 3	Click Save button to add/update supplier record	UIE
Event 4	System validates inputs and updates or saves record	DP
Event 5	Display success or error message	OP

User 6 Goal 6

Track Pending Supplier Deliveries

Workflow:

Event #	Description	Type
Event 1	Click “Track Pending Deliveries” from dashboard	UIE
Event 2	System loads all supplier orders from the saved records	DP
Event 3	System loads delivery status from Inventory Manager	DP
Event 4	System compares delivered vs pending orders	DP
Event 5	Display pending orders in text area	OP

User 6 Goal 7

Generate Procurement Report

Workflow:

Event #	Description	Type
Event 1	Click “Generate Procurement Report” from dashboard	UIE

Event 2	System loads all supplier orders from saved records	DP
Event 3	Display orders in text area	OP
Event 4	Click “Download Report” to export as PDF	UIE
Event 5	System generates PDF with order details	DP
Event 6	Show confirmation message for successful download	OP

User 6 Goal 8

View Supplier Contact Information

Workflow:

Event #	Description	Type
Event 1	Click “View Supplier Contact” from dashboard	UIE
Event 2	Enter Supplier ID (from a manual sheet or report)	UID
Event 3	Click “Search”	UIE
Event 4	System searches supplier records file	DP
Event 5	If found, display supplier details in text area	OP
Event 6	Show success message (Supplier found) or error if not found	OP

2312026 – Tanjil Sarkar

User 7: **HR Manager**

User 7 Goal 1

Recruit Staff: The HR Manager can recruit a new staff member by entering their name, selecting department & position, and saving employee information into the system.

Workflow:

Event #	Description	Type
Event 1	HR Manager opens the **Recruit Staff** page → the system loads all departments and positions into dropdown lists.	UIE
Event 2	HR enters **Full Name** and selects **Department** + **Position** from dropdowns.	UID
Event 3	HR clicks **Add Employee**.	UIE
Event 4	System validates that all fields are filled; if missing → shows **“Please fill in all fields.”**	VL / OP
Event 5	If valid, the system generates a **unique ID**, creates an **Employee object**, and adds it to `employeeList`.	DP
Event 6	The system shows **“Employee added successfully!”** and clears all input fields.	OP
Event 7	HR clicks **Save to File**, and the system writes the updated employee list into **employee.bin**.	UIE / DP/ OP

User 7 Goal 2

Remove Employee: *The HR Manager can remove an existing employee from the system.They can load all employee records, select the employee they want to remove from the table, and the system will update the employee list and save the changes to the file.*

Workflow:

Event #	Description	Type
Event 1	HR Manager opens the **Remove Employee** page → system sets up table columns (ID, Name, Department, Position).	UIE
Event 2	HR clicks **Load Employees**, and the system reads `employee.bin` and displays all employees in the table.	UIE/ DP/ OP
Event 3	HR selects an employee row from the table.	UID
Event 4	HR clicks **Remove Employee**.	UIE
Event 5	System checks if an employee is selected; if none → shows **“Please select an employee to remove.”**	VL/ OP
Event 6	If valid, the system removes the selected employee from `employeeList`, updates the table, and saves the new list back to `employee.bin`.	DP
Event 7	The system displays **“Employee removed successfully!”**	CP

2312026 – Tanjil Sarkar

User 7 Goal 3

Update Employee Info: The HR Manager can search for an employee by name, view their existing department and position, update these details, and save the updated information back into the system file.

Workflow:

Event #	Description	Type
Event 1	HR Manager opens the page → system loads all departments & positions into dropdown lists, and table columns are initialized.	UIE
Event 2	HR clicks **Load Employees**, and the system reads `employee.bin` and displays all employees in the table.	UIE / DP / OP
Event 3	HR types a name in the Search **Name** field and clicks **Search**.	UID / UIE
Event 4	The system checks if the search field is empty → if empty, shows **"Enter a name to search."**	VL / OP
Event 5	If the name matches, the system selects that employee, fills in Department & Position dropdowns, and displays **"Employee found."**	DP / OP
Event 6	HR updates Department/Position and clicks **Update** → system validates selections.	UID / UIE / VL
Event 7	System updates the employee object, refreshes the table, saves updated list to `employee.bin`, and shows **"Employee updated successfully!"**	DP / OP

User 7 Goal 4

Approve Leave Requests: The HR Manager can load all pending leave requests, view employee names, leave types, dates, and current status, then select any request and either approve or reject it. The updated status is saved back into the system file.

Workflow:

Event #	Description	Type
Event 1	HR opens the page → system initializes the table columns (Employee Name, Leave Type, Start Date, End Date, Status).	UIE
Event 2	HR clicks **Load Requests**, system reads `leaveRequests.bin` and displays all leave requests in the table.	UIE / DP / OP
Event 3	HR selects a leave request from the table.	UIE
Event 4	HR clicks **Approve** or **Reject** → system checks if a request is selected.	UIE/ VL
Event 5	If no item selected → system displays message **"Select a leave request to approve/reject.**	OP
Event 6	If selected → system updates the status to *Approved* or *Rejected*.	DP
Event 7	System saves updated list to `leaveRequests.bin`, refreshes table, and shows confirmation message.	DP/ OP

User 7 Goal 5

Manage Employee Attendance: The HR Manager can load employees from the system, select an employee, mark their attendance for the current date as Present, Absent, or Leave, view all attendance records, save the data to a file, and clear the attendance table when needed.

Workflow:

Event #	Description	Type
Event 1	HR Manager opens the Attendance page → system initializes employee and attendance table columns and loads status options (“Present, Absent, Leave”) into the dropdown.	UIE
Event 2	HR clicks **Load Employees** → system reads `employee.bin` and displays all employees in the employee table.	UIE / DP / OP
Event 3	HR selects an employee from the table and selects a status from the dropdown.	UID
Event 4	HR clicks **Mark Attendance** → system checks if an employee and status are selected. If not, it shows validation messages: “Select an employee first.” or “Select attendance status.”	UIE / VL / OP
Event 5	If valid, the system creates a new Attendance object with employeeName, current date, and selected status, adds it to `attendanceList`, and updates the attendance table.	DP/OP
Event 6	HR clicks **Save Attendance** → system writes `attendanceList` into `attendance.bin` and shows **“Attendance saved successfully!”**	UIE / DP / OP
Event 7	HR clicks **Clear Attendance** → system clears the attendance table and resets `attendanceList`, showing **“Attendance table cleared.”**	UIE / DP / OP

User 7 Goal 6

Manage Payroll: The HR Manager can load employees, view their current salary, update it with a new value, validate input, save the updated salary back to the system file, and see a confirmation message.

Workflow:

Event #	Description	Type
Event 1	HR Manager opens the Payroll page → system initializes table columns (ID, Name, Department, Position, Salary) and sets up a listener to display selected employee’s salary in the text field.	UIE
Event 2	HR clicks **Load Employees** → system reads `employee.bin` and displays all employees in the table.	UIE / DP / OP
Event 3	HR selects an employee from the table → system shows the selected employee’s current salary in the salary text field.	UIE / OP
Event 4	HR enters a new salary in the text field.	UID
Event 5	HR clicks **Update Salary** → system validates input. If empty or non-numeric → shows **“Enter a valid number for salary.”**	UIE / VL / OP

Event 6	If valid → system updates the selected employee's salary in `employeeList`, refreshes the employee table, saves updated list to `employee.bin`, clears selection and text field.	DP / OP
Event 7	The system displays **"Salary updated successfully!"** confirming the update	OP

User 7 Goal 7

Leave Request: The HR Manager can submit leave requests for employees by entering the employee's name, selecting a leave type, specifying start and end dates, view all requests, save them to a file, and clear the leave requests table when needed.

Workflow:

Event #	Description	Type
Event 1	HR Manager opens the Leave Request page → system initializes table columns (Employee Name, Leave Type, Start Date, End Date, Status) and populates leave type dropdown.	UIE
Event 2	HR clicks **Load Requests** → system reads `leaveRequests.bin` and displays all leave requests in the table.	UIE / DP / OP
Event 3	HR enters employee name, selects leave type, start date, and end date.	UID
Event 4	HR clicks **Submit Request** → system checks if all fields are filled → if not, shows **"Fill all fields before submitting."**	UIE / VL / OP
Event 5	If valid, the system creates a new LeaveRequest object with status "Pending", adds it to `leaveRequests`, updates the leave table, and saves to `leaveRequests.bin`.	DP / OP
Event 6	System clears input fields (employee name, leave type, start and end dates) after submission.	DP / OP
Event 7	HR clicks **Clear Table** → system clears all leave requests from the table and shows **"Table cleared."**	UIE / DP / OP

User 7 Goal 8

View HR Summary Report: The HR Manager can load all employee records from employee.bin, view them in a table, and export the full HR Report as a PDF. The system validates data, loads employee information, generates the PDF with all details, and shows confirmation messages.

Workflow:

Event #	Description	Type
Event 1	HR Manager opens the HR Report page → TableView columns are initialized (ID, Name, Department, Position, Salary).	UIE
Event 2	HR Manager clicks **Load Report** → system attempts to read employee.bin and populate the employeeTable.	UID / OP
Event 3	If the file is missing/corrupted → system displays “Could not load employee data.” Otherwise, the table loads successfully.	VL / OP / UIE
Event 4	HR Manager clicks **Export to PDF** → system checks whether employeeList is empty.	UID / VL
Event 5	If empty → shows “No employee data to export!”. If valid → FileChooser opens and the system generates the HR Report PDF containing all employee details.	VL / OP
Event 6	After successful PDF creation → system displays “PDF generated successfully!” (or error message if failed).	OP / UIE

User 8: **Customer Support Staff**

User 8 Goal 1

Add New Complaint: The Customer Support Staff can submit a new complaint by entering the customer’s name, complaint title, and description. The complaint is saved to `complaints.txt` with a default status of “Pending.”

Workflow:

Event #	Description	Type
Event 1	Customer Support Staff opens the Add Complaint page → system displays fields for Customer Name, Complaint Title, and Description.	UIE
Event 2	Staff types customer name, complaint title, and description into the respective fields.	UID
Event 3	Staff clicks **Submit Complaint** → system validates that all fields are filled.	UIE / VL
Event 4	If any field is empty, the system displays **“Please fill all fields.”**	VL/ OP
Event 5	If all fields are filled, the system creates a Complaint object with default status “Pending.”	DP
Event 6	The system appends the complaint to `complaints.txt` and shows **“Complaint submitted successfully!”**	DP/ OP
Event 7	System clears all input fields (Customer Name, Complaint Title, Description) for the next entry.	DP/ OP

User 8 Goal 2

Update Complaint Info: The Customer Support Staff can search for complaints by customer name, view all matching complaints in a table, and mark a selected complaint as “Solved.” The updates are saved back to `complaints.txt`.

Workflow:

Event #	Description	Type
Event 1	Customer Support Staff opens the Update Complaint page → system initializes table columns for Complaint Title, Description, and Status.	UIE
Event 2	Staff types a customer name into the search field.	UID
Event 3	Staff clicks **Search Complaint** → system validates the input.	UIE / VL
Event 4	If the search field is empty, the system displays **"Please enter a customer name."**	VL / OP
Event 5	If the name is valid, the system reads `complaints.txt`, filters complaints matching the customer name, and displays them in the table.	DP / OP
Event 6	Staff selects a complaint and clicks **Mark as Solved** → system updates the complaint's status to "Solved" in memory and writes all lines back to `complaints.txt`.	UID / DP / OP
Event 7	System refreshes the table with updated complaint statuses and displays **"Complaint marked as Solved."**	DP / OP

User 8 Goal 3

Close Complaint: The Customer Support Staff can view all pending complaints in a table, select a complaint, and close it. The system updates `complaints.txt` to remove the closed complaint from the pending list and refreshes the table.

Workflow:

Event #	Description	Type
Event 1	Customer Support Staff opens the Close Complaint page → system initializes table columns for Customer Name and Status.	UIE
Event 2	Staff clicks **Load Complaints** → system reads `complaints.txt` and filters out complaints that are already closed.	UIE / DP / OP
Event 3	The system populates the table with pending complaints.	DP / OP
Event 4	If no pending complaints exist, the system displays **"No pending complaints."**	VL / OP
Event 5	Staff selects a complaint from the table.	UID
Event 6	Staff clicks **Close Complaint** → system validates selection.	UIE / VL
Event 7	The system removes the selected complaint from `complaints.txt`, refreshes the table, and displays **"Complaint closed successfully!"**	DP / OP

User 8 Goal 4

Check Customer Order Status: *Staff can check a customer's order or delivery progress from the system.*

Workflow:

Event #	Description	Type
Event 1	Click “Check Order Status”	UIE
Event 2	Enter customer ID or order number	UID
Event 3	Fetch order details	DP
Event 4	Display order progress	OP
Event 5	Validate order ID	VL

User 8 Goal 5

Register New Membership Request: The Customer Support Staff can register a new membership request by entering the customer’s name, contact information, and optional notes. The system saves the request as **Pending** in `membership_requests.txt` and clears the form..

Workflow:

Event #	Description	Type
Event 1	Customer Support Staff opens the Register Membership Request page → system initializes input fields and message labels.	UIE
Event 2	Staff enters customer names, contact info, and optional notes.	UID
Event 3	Staff clicks **Submit Request** → system validates that customer name and contact info are not empty.	UIE / VL
Event 4	If the required fields are empty, the system displays **“Please enter customer name and contact info.”**	VL / OP
Event 5	If validation passes, the system creates a new membership request record with status **Pending**.	DP / OP
Event 6	The system writes the new request to `membership_requests.txt`.	DP / OP
Event 7	The system clears all input fields and displays **“Request submitted successfully!”**	UIE / OP

User 8 Goal 6

Pending Complaint Reminder: The Customer Support Staff can view all pending complaints by loading them from `complaints.txt`. Only complaints that are not **Closed** are displayed in a table showing customer name, complaint title, and status.

Workflow:

Event #	Description	Type
Event 1	Customer Support Staff opens the Pending Complaint Reminder page → table columns are initialized.	UIE
Event 2	Staff clicks **Load Pending Complaints** → system clears previous data.	UIE / OP
Event 3	The system reads `complaints.txt` line by line.	DP

2312026 - Tanjil Sarkar

Event 4	For each line, the system splits into customer name, title, description, and status.	DP
Event 5	The system filters complaints where status is **not Closed**.	DP / OP
Event 6	System adds filtered complaints to table (`tblPendingComplaints`).	DP / UIE
Event 7	The system displays the message **“Loaded pending complaints.”**	OP / UIE

User 8 Goal 7

Submit Customer Feedback: The Customer Support Staff can submit feedback from a customer by entering the customer’s name and feedback content. The system saves the feedback to `feedback.txt` and confirms submission

Workflow:

Event #	Description	Type
Event 1	Customer Support Staff opens the Feedback page → text areas and labels are initialized.	UIE
Event 2	Staff enters customer name and feedback into `txtCustomerName` and `txtFeedback`.	UID / UIE
Event 3	Staff clicks **Submit** → system checks if either field is empty.	UID / VL
Event 4	If any field is empty → system displays **“Please enter customer name and feedback.”**	VL / OP
Event 5	If both fields are filled → system writes feedback to `feedback.txt` in the format `CustomerName	UIE
Event 6	The system clears the text fields after saving.	UIE / OP
Event 7	The system displays **“Feedback submitted successfully!”**	OP / UIE

User 8 Goal 8

Generate Support Report: The Customer Support Staff can load all complaints from **complaints.txt** and view customer names with their complaint status in a TableView.

Workflow:

Event #	Description	Type
Event 1	Customer Support Staff opens the Support Report page → Table columns (Customer Name, Status) are initialized and bound.	UIE
Event 2	Staff clicks **Load Complaints** → system opens complaints.txt to read complaint records.	UID / OP
Event 3	The system adds each complaint as a `SupportReport` object into the TableView’s ObservableList.	OP / UIE
Event 4	If no complaints exist → system shows “No complaints found.”	VL / UIE
Event 5	If data exists → system displays “Complaints loaded successfully.”	OP / UIE

2312026 – Tanjil Sarkar